Antonin Vidon

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Work Experience

Jan 2023 – Present
Staff Machine Learning Engineer
Hyperscience, New York, USA
A SaaS and on-prem document AI platform built on proprietary as well as fine-tuned open-source models.

Drove development of a Vision Language Model (VLM) pipeline for structured data extraction from document streams.

- Cut cost per document by up to 10× on self-hosted VLM inference: re-architected the service around SGLang as a multi-threaded pipeline with continuous batching, and built a VRAM cost model that auto-tunes batch size, KV cache and thread count per machine to maximize throughput within GPU and host memory limits.
- Led cross-team delivery of a human-in-the-loop system that holds automated VLM predictions to a customer-set accuracy target and speeds up review of the rest: derived per-field confidence from log-probs or a small surrogate via teacher forcing, fit thresholds to that target, and recomputed locations based on attention scores for unconfident fields.
- Reached up to 95% automation at 98% accuracy from only a handful of annotated samples, building end-to-end training (LoRA, layer freezing), few-shot and evaluation pipelines with VLMs and layout-aware LLMs.
- Enabled cross-functional teams to deploy, monitor and customize VLMs without ML-team support: built per-pod Grafana observability of model state, errors and serving performance on AWS and GCP, plus guides on hardware requirements, fine-tuning and prompting.
- Mentored 4 NYU students in training an OCR-free vision transformer for document Q&A, from preprocessing to evaluation, on 200k synthetic Q&A pairs over 70k multilingual Common Crawl PDFs.

Jun 2022 – Aug 2022
Machine Learning Research Intern
Hyperscience, New York, USA

- Led end-to-end research project to reconstruct human reading order from OCR output on complex documents (e.g. multi-column layouts, tables), providing downstream models with coherent context.
- Developed clustering-based methods and trained a Faster R-CNN layout detector, directing an internal team on annotation and building the training and evaluation pipeline; integrated the solution into production, improving downstream accuracy by +3.9% for auto-annotation and +3.8% for extraction.

Mar 2021 – Aug 2021
RL Research Scientist Intern
Huawei, London, United Kingdom

- Extended Deep Imitative Models to remove the need for human demonstrations, training a deep generative model on "expert-like" trajectories self-collected by a recurrent A2C agent from raw pixels in first-person navigation (VizDoom).
- Planned trajectories by optimizing a posterior combining a normalizing-flow imitation prior with a flexible task-specific goal likelihood, yielding paths that adapted to surface friction; the prior also transferred obstacle avoidance to unseen maps, cutting mean episode length by 30% vs. a prior trained without obstacles.

Jun 2020 – Aug 2020
Associate Consultant Intern
Bain & Company, Paris, France

- Co-designed a $2B, 3-year cost-reduction program for a large European manufacturer in a client-facing role, and analyzed market potential for wind turbine manufacturers from financial datasets.

Education

Dec 2022
MS in Data Science
Columbia University, New York, USA

Aug 2021
MS in Applied Mathematics
École Polytechnique, Palaiseau, France

- Capstone ranked 1st/112 ("Excellence prize"): Integration of physical models into voxel-based video games.

Blog Posts

Jul 2025 — *"Out-of-the-Box to State-of-the-Art: How Vision Language Models Are Transforming Document Processing"*

Nov 2024 — *"Managing Randomness During Training in Machine Learning"*

Skills

- **Stack:** Python, PyTorch, SGLang, vLLM, AWS, GCP
- **Languages:** English, French (native), German (intermediate)